

SUSMA R

DATA SCIENTIST / ML ENGINEER / AI ENGINEER

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SUMMARY

A highly motivated and results-driven Decision Science graduate with strong analytical skills and a solid foundation in machine learning and artificial intelligence. Proficient in Python, data analysis, and various ML frameworks, eager to apply knowledge in real-world problem-solving and contribute to innovative AI projects.

WORK EXPERIENCE

Software Engineer (AI), RedBlackTree Technologies Pvt Ltd (Chennai) Jan 2025 - Present

- Developed and implemented an AI-powered invoice processing system using **LangGraph** and **LLMs**.
- Automated data extraction, supplier, account, store assignment, and manual journal entry determination.
- Integrated with a **PostgreSQL** database for robust data management and workflow state persistence.
- Significantly reduced manual effort and improved accuracy in financial document processing.

Tech Stack: Python, LangGraph, SQLAlchemy, PostgreSQL, and Google Gemini.

Data Scientist Internship, Bosch Global Software Technologies (Bangalore) Jan 2024 - May 2024

- Designed and implemented an **End-to-End Robustness Evaluation Pipeline** to assess AI/ML model performance under Adversarial and real-world conditions.
- Utilized **GANs** for synthetic data generation, Adversarial Testing, and Randomized Smoothing to evaluate model stability and resilience.
- Integrated Out-of-Distribution Detection and Counterfactual Explanation techniques to assess robustness and interpretability.
- Built and deployed scalable **FastAPI** services for real-time evaluation and visualization of robustness metrics.

Tech Stack: Python, FastAPI, Machine Learning, GANs, Responsible AI, API Development

Project Trainee, Bosch Global Software Technologies (Coimbatore) Jul 2022 - Nov 2022

- Automated data collection from the **JIRA** server for real-time ticket status tracking and key metrics, ensuring streamlined reporting.
- Developed an interactive **RNM Metrics Dashboard** using Plotly and Dash to visualize ticket priorities, productivity, and quality, enhancing decision-making.

Tech Stack: Python, JIRA, Plotly, Dash

PROJECTS

Smart AI Interview Evaluation and Coaching Platform

Developed an AI-powered interview analysis system to process transcripts, detect bias, and generate structured STAR feedback. Implemented a multi-agent CrewAI architecture with specialized agents for skills mapping, coaching, and summary generation. Built pipelines for file processing, data validation, report generation, and secure API integration. Created a scalable, production-ready solution converting raw interview data into actionable insights for HR teams.

Tech Stack : Python, Streamlit, Mistral AI, CrewAI

Legal Information Retrieval System

Developed an AI-powered system using Python and OpenAI to translate user queries into relevant legal information, integrating Llama Index, Chainlit, and a local vector database. Implemented tools for extracting key legal concepts and identifying corresponding clauses in updated legal texts, ensuring precision and relevance.

Tech Stack : Python, OpenAI, Llama Index, Chainlit, Local Vector Database

Real-Time Squat Tracking and Analysis System

This project implements a real-time squat tracking system with under 100ms latency using Flask, OpenCV, and MediaPipe’s 33 keypoint pose estimation. It processes 1080p video, calculates joint angles (hip, knee, ankle), and tracks squats at 30 FPS. Keypoints and angles are displayed live, enabling advanced movement analysis for fitness applications.

Tech Stack : Python, HTML, Flask (Python), OpenCV, MediaPipe

SKILLS

Areas of Interest	Data Science, Machine Learning, Deep Learning, NLP, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), GANs, Statistics, Database Management System
Programming Languages	Python, SQL
Frameworks and Platforms	Tensorflow, Pytorch, Scikit-learn, Langchain, OpenCV, Flask, FastAPI
Database	MySQL, MongoDB, VectorDB, PostgreSQL
Tools	PowerBI, Tableau

EDUCATION

M.Sc. Decision and Computing Sciences(Integrated), Coimbatore Institute of Technology 2024 (CGPA: 8.69)

PUBLICATIONS

- “A Real Time Web-Based Flask Application For Detection Of Facial Emotions Using Deep Learning Model” (Apr. 2023). In: *JASC*.
- “Mushroom Classification To Detect Their Edibility using ML Algorithms and Rapid Miner” (Apr. 2023). In: *IRJMETs*.
- “Comprehensive Framework for Robustness Evaluation on Numeric Data Classification” (June 2024). In: *ICCCNT 2024 held at IIT-Mandi, IEEE Electronics Packaging Society*.